Specimen Calculations:

Difference in divisions = 10

Angular Deviation = Difference in Divisions * Scale = 10

Linear Deviation

= Tan (Angular Deviation in minutes / 60 + Angular Deviation in Seconds / 3

= tan(0/60 + 10/3600) x10^5 = 4.84 μm

Graph

Plot a graph of Position v/s linear deviation in microns

Inference:

(Which areas of the surface deviate most from flatness, and what could be the possible cause)

Maximum deviation from flatness usually occurs near the su
edges compared to the center. Possible cause include uneven suppo
improper leveling or thermal distortion during measurement with a

Application Question: autocollimator.

1. Collect the readings of other batches.
2. For each position, calculate the mean and range (or standard deviation) readings.
3. Discuss the following:
   a) Are the readings consistent across batches?
   b) Which shows more variation — repeated measurements by the same (repeatability) or measurements by different people (reproducibility)?
4. Explain why it is important to check repeatability and reproducibility in measurements.?

Particulars	Marks Awarded	Marks Obtained
1. Conceptual Understanding	04	
2. Conduction of Experiments	04	
3. Write up & Performance Accuracy	04	
4. Inference	04	
5. Viva	04	
Total	20	
Signature with Date		

R.V. COLLEGE OF ENGINEERING®

OBSERVATION / DATA SHEET

Date 27/10/25 Name Sudhanva · Sudha

Dept./Lab DM Class IEM Expt./No. E03

Title Auto collimator

Batch 1:

Sl NO	Length of autocollimator bank l(mm)	Position of reflector	Autocollimator Reading		θ in seconds	Linear displacement in micron(L)	Error in micron (+) or (-)
			I_{SR}	M_R			
1	0-100	ab	1000	10,00	0	0	0
2	100-200	bc	10,00	9,106	14	6.7873	-6.78
3	200-300	cd	10,00	10,06	06	2.9088	2.90
4	300-400	de	10,00	10,16	16	7.7570	7.75
5	400-500	ef	10,00	10,22	22	10.6659	10.66
6	500-600	fg	10,00	10,13	13	6.3025	6.30
7	600-700	gh	10,00	10,01	01	0.4848	0.48
8	700-800	hi	10,00	9,112	08	3.8785	-3.87
9	800-900	ij	10,00	10,18	18	8.7266	8.72

E03

Signature of Teacher Incharge

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OBSERVATION / DATA SHEET

Date 27/10/25 Name Sudhanva Geetha

Dept./Lab DM Class IEM Expt./No. E03

Title Autocollimator.

Batch: 3

Sl No	Length of offset taken l(mm)	Position of Reflector	Autocollimator reading		'd' in seconds	Linear displacement in microns (s)	Error in microns (+) or (-)
			IR	MR			
1	0-100	ab	10,00	10,00	0	0	0
2	100-200	bc	10,00	10,41	41	19.87	19.87
3	200-300	cd	10,00	10, 10	10	4.8	4.8
4	300-400	de	10,00	10,45	45	21.8	21.8
5	400-500	ef	10,00	10,35	35	16.9	16.9
6	500-600	fg	10,00	10,24	24	11.6	11.6
7	600-700	gh	10,00	10,37	37	17.9	17.9
8	700-800	hi	10,00	10,13	13	6.30	6.30
9	800-900	ij	10,00	10,46	46	2.23	2.23

Signature of
Teacher Incharge

Calculating mean for every position.

1) ab = 0/4 = 0

2) bc = (4.84 + 6.78 + 2.42 + 19.87)/4 = 8.47

3) cd = (16.48 + 2.90 + 1.93 + 4.8)/4 = 6.52

4) de = (4.36 + 7.75 + 8.72 + 21.8)/4 = 10.65

5) ef = (15.99 + 10.66 + 9.69 + 16.9)/4 = 13.31

6) fg = (1.93 + 6.30 + 1.93 + 11.6)/4 = 5.44

7) gh = (29.08 + 0.48 + 8.24 + 17.9)/4 = 13.92

8) hi = (18.42 + 3.87 + 4.84 + 6.30)/4 = 8.35

9) ij = (35.39 + 8.72 + 6.30 + 2.23)/4 = 13.16

Calculating range for every position.

1) ab = 0 - 0 = 0

2) bc = 19.87 - 2.42 = 17.45

3) 14.55

4) de = 17.44

5) ef = 7.21

6) fg = 9.67

7) gh = 28.6

8) hi = 14.55

9) ij = 33.16

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OBSERVATION / DATA SHEET

Date 27/10/25 Name Sudhanva. Seveka

Dept./Lab DM Class IEM Expt./No. E03

Title Auto collimator

3)a) The readings across d/f batches are not consistent and show minor variations from batch to batch due to :-

* Repositioning of the reflector/mirror.
* Environmental effects like temp. and vibrations.
* Instrument resolution.

b) Reproducibility shows more variation because :-

* Diff operators align the mirror differently.
* Reading and forming errors may from person to person.
* Diff in handling instrumental setup.

Here, repeatability is better than reproducibility.

4) During repeatability and reproducibility is essential in quality measurements because :-

* Ensures reliability of measurements.
* Checks instrument and plant operations.

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* Improve confidence in objective valuation.
* Support quality control and standardization.
* Reduce rejection and rework.

Thermometer reading
Thermometer Reading
Thermometer Reading
Thermometer Reading
Thermometer Reading
Thermometer Reading

RV College of Engineering®, Bangalore – 59
(Autonomous Institution affiliated to VTU, Belgaum)
Department of Industrial Engineering and Management

Digital Metrology - IM235AI
OBSERVATION / DATA SHEET

Observations:

- Range of thermometer = 0-100°c
- Least Count of thermometer = 1°c
- Least Count of Digital temperature indicator = 0.1°C
- Thermocouple material = Chromel-Alumel
- Calibration range Selected = 50° to 68°c in steps of 2°c

Tabulation:

Sl. No.	Thermometer Reading T_T ^{0}C		Thermocouple reading T_{Tc} ^{0} C		% Error = $\frac{(T_{TC} - T_T)}{T_T} x100$	
	Heating	Cooling	Heating	Cooling	Heating (-)	Cooling (-)
1	50	68	44.2	66	-11.6	-2.94
2	52	66	46.0	65	-11.53	-1.51
3	54	64	47.8	62.6	-11.48	-2.18
4	56	62	50.2	60.2	-10.35	-2.90
5	58	60	52.3	58.5	-9.82	-2.5
6	60	58	54.5	57.1	-9.16	-1.55
7	62	56	57.2	55	-7.74	-1.78
8	64	54	58.8	52.6	-8.12	-2.59
9	66	52	61.2	50.5	-7.27	-2.88
10	68	50	63.5	48	-6.61	-4.0

Signature of staff in-charge

Specimen calculation:

For heating,

Error = $T_T = 50 \quad T_{TC} = 44.2$

% Error = $T_{TC} - T_T = -5.8$

$= \frac{-5.8}{50} \times 100 = -11.6\%$

For cooling,

Error = $T_T = 68 \quad T_{TC} = 66$

$= T_{TC} - T_T = -2$

% Error = $\frac{-2}{68} \times 100 = -2.94\%$

Graphs:

1. Thermometer reading v/s Thermocouple Reading
2. Thermometer reading v/s % Error

Inference: Thermocouple is a temp. measuring device. The working principle is based on seebeck effect. They are made of metals are joined at one end where one end is exposed to cold and other to hot.

1. Using your calibration readings, calculate the hysteresis error at each temperature.

$$\text{Hysteresis Error} = | TTC_{\text{heating}} - TTC_{\text{cooling}} |$$

2. Tabulate: Temperature, Heating Reading, Cooling Reading, Hysteresis Error.
3. Plot Hysteresis Error vs Temperature.
4. Answer briefly:
   a) Which temperature range shows the highest hysteresis?
   b) How can hysteresis affect industrial temperature measurements?
   c) Suggest ways to reduce hysteresis error.

Particulars	Marks Awarded	Marks Obtained
1. Conceptual Understanding	04	
2. Conduction of Experiments	04	
3. Write up & Performance Accuracy	04	
4. Inference	04	
5. Viva	04	
Total	**20**	
Signature with Date		

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OBSERVATION / DATA SHEET

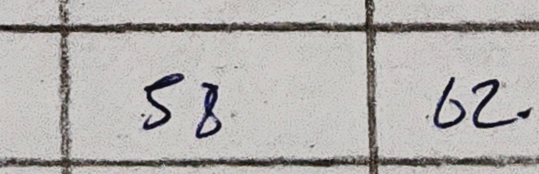
Date 13/10/25 Name Sudharsan Jesitha

Dept./Lab DM Class IEM Expt./No. E06

Title Thermocouple.

Hysterisis error calculation :-

Sl No	T_T°C Reading		T_{Tc}°C Reading		Hysterisis Error
	Heating	Cooling	Heating	Cooling	
1	50	68	52.9	72.5	0
2	52	66	53.6	70.4	2.5
3	54	64	54.5	68.1	3.3
4	56	62	56.8	66.4	2.9
5	58	60	60.2	64.3	1.1
6	60	58	62.2	61.3	2.1
7	62	56	65.2	59.7	1.2
8	64	54	67.2	57.8	0.9
9	66	52	68.8	56.1	1.6
10	68	50	70.9	52.9	1.6



E06

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4) a) The 50°C - 60°C range shows the widest hysteresis.

* This range has maximum gap blw heating and cooling graph.
curves in the.

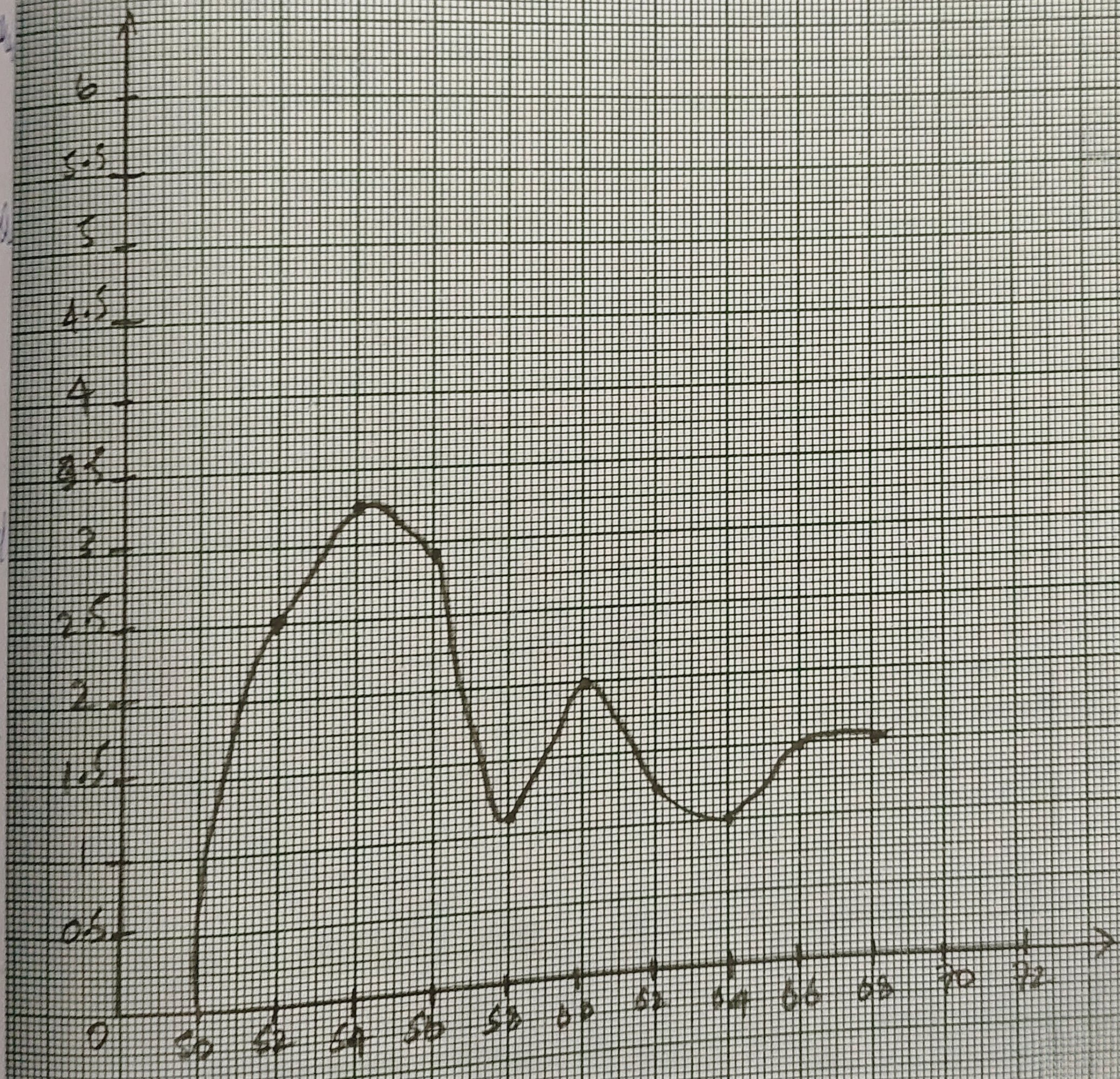
b) * It causes dct readings at two same temp during heating and cooling.
* This leads to errors in power control and reduced accuracy.

c) * Use sensor with low hysteresis characteristics.
* Ensure proper calibration and avoid rapid temp changes.
* Maintain stable and stable operating conditions.

Scale:

Y axis: 1 unit = 1°C

X axis: 1 unit = 0.5°C



Formula used:

Actual displacement $A = V_o - V_i$

% Error, $e_i = \{(I_i - A_i) * 100\} / A_i$

Where,

I_i = Indicated displacement,

A_i =Actual displacement

Least count Calculation:

Least count of micrometer gauge :-

Main scale division (MSD) = 1 mm

No. of circular scale division = 50

$$\text{Least count} = \frac{1\ \text{MSD}}{\text{No. of circular scale division}}$$

$$= \frac{1}{50}$$

$$LC = 0.02\ mm$$

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Department of Industrial Engineering and Management

Digital Metrology - IM235AI

OBSERVATION / DATA SHEET

Observations:

Equipment specifications:

1. Linearity range of LVDT = ±20mm
2. Least count of the displacement indicator= 0.01mm
3. Least count of the Vernier height gauge = 0.02mm

Initial set up:

1. Initial reading of the displacement indicator = 0
2. Corresponding reading of the Vernier height gauge (Vo) = 30.64mm

Tabulation:

Sl. No	Backward traverse:		Forward traverse:	
	Height gauge Reading V_i (mm)	Displacement Indicator reading I_i (mm)	Height gauge Reading V_i (mm)	Displacement Indicator reading I_i (mm)
1	32.64	-2.01	28.64	2.07
2	34.64	-4.01	26.64	4.09
3	36.64	-6.06	24.64	6.09
4	38.64	-8.07	22.64	8.11
5	40.64	-10.05	20.64	10.15
6	42.64	-12.04	18.64	12.17
7	44.64	-14.03	16.64	14.23
8	46.64	-16.05	14.64	16.19
9	48.64	-18.01	12.64	18.16
10	50.64	-19.99	10.64	19.99

Signature of the staff in-charge

Tabulation and calculations

Initial reading of the displacement indicator = 0

Corresponding reading of the Vernier height gauge (Vo) = 30.64mm

A. Backward Traverse

Sl. No	Height gauge Reading V_i (mm)	Displacement Indicator reading I_i (mm)	Actual displacement, A_i (mm)	% Error,
1	32.64	-2.01	2	-0.50
2	34.64	-4.01	4	-0.25
3	36.64	-6.06	6	-1.00
4	38.64	-8.07	8	-0.88
5	40.64	-10.05	10	-0.50
6	~~42~~.64	-12.04	12	-0.33
7	~~44~~.64	-14.03	14	-0.21
8	46.64	-16.05	16	-0.31
9	48.64	-18.01	18	-0.06
10	50.64	-19.99	20	-0.05

B. Forward traverse:

Sl. No	Height gauge Reading V_i (mm)	Displacement Indicator reading I_i (mm)	Actual displacement, A_i (mm)	% Error,
1	28.64	2.07	2	3.50
2	26.64	4.09	4	2.25
3	24.64	6.09	6	1.50
4	22.64	8.11	8	1.38
5	20.64	10.15	10	1.50
6	18.64	12.17	12	1.42
7	16.64	14.23	14	1.64
8	14.64	16.19	16	1.19
9	12.64	18.16	18	0.89
10	10.64	19.99	20	-0.05

Specimen Calculations:

For, backward traverse (1st reading)

Actual displacement, $A_1 = V_0 - V_1$ = 32.64 - 30.64 = 2mm

Indicated displacement, I_1 = 2.01mm

% Error = $\{(I_i - A_i) * 100\} / A_i$ = $\frac{(2.01-2) \times 100}{2}$

= 0.50%

For, forward traverse (1st reading)

Actual displacement, $A_1 = V_0 - V_1$ = 30.64 - 28.64 = 2mm

Indicated displacement, I_1 = 2.07

% Error = $\{(I_i - A_i) * 100\} / A_i$ = $\frac{(2.07-2) \times 100}{2}$

= 3.5%

Graphs:

1. Actual displacement v/s Displacement Indicator reading
2. Actual displacement v/s % Error

Inference:

The calibration results show the LVDT ~~results~~ shows that the LVDT output varies linearly with displacement within its operating range. A small hysteresis is present b/w the forward and backward travels, with maximum hysteresis occurring in the mid displacement range, indicating good sensitivity and acceptable accuracy of LVDT.

Screw 2:-

Majos ϕ

Position	Rp(mm)	R(mm)	Rang	Rs-R
I	5.1200	18.4620	18.4602	13.3402
II		18.4556		
III		18.4630		

minos ϕ

Position	Rs(mm)	R(mm)	Rang	Rs-R
I	12.34	21.2804	21.2746	8.9346
II		21.2700		
III		21.2734		

Effective ϕ

Position	Rf(mm)	R(mm)	Rang	Rf-R
I	8:1100	18.6902	18.6980	10.5880
II		18.7024		
III		18.7014		

Batch 2:

Sl NO	Length of substrate bore l(mm)	Position of the substrate	Autocollimator reading		(α) in Seconds	Linear Displacement in micron (h)	Error (+)
			I_{S2}	M_R			
1	0-100	ab	10,00	10,00	0	0	0
2	100-200	bc	10,00	10,05	5	2.42	2.42
3	200-300	cd	10,00	10,04	4	1.93	1.93
4	300-400	de	10,00	9,102	18	8.72	-8.72
5	400-500	ef	10,00	10,20	20	9.64	9.64
6	500-600	fg	10,00	9,116	4	1.93	-1.93
7	600-700	gh	10,00	10,17	17	8.24	8.24
8	700-800	hij	10,00	9,110	10	4.84	-4.84
9	800-900	ij	10,00	10,13	13	6.30	6.30